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Impact of methadone treatment initiated in jail on subsequent arrest

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ABSTRACT

Background: There are limited data from randomized trials about the impact of starting methadone treatment in jail on subsequent arrest after release for adults with opioid use disorder (OUD).
Methods: Official arrest records were obtained for 212 participants with OUD who were enrolled in a three-group randomized controlled trial of initiating methadone treatment in jail either with or without patient navigation *vs.* enhanced treatment-as-usual in Baltimore, Maryland. Participants treated for opioid withdrawal in jail were assigned to: 1) interim methadone (IM) with patient navigation (PN; IM + PN); 2) IM without PN (IM); or 3) enhanced treatment-as-usual (ETAU). Participants in both IM groups were able to continue treatment at a community-based methadone treatment program with counseling upon release, while ETAU participants received overdose information and a city-wide treatment assessment/referral number. Likelihood of arrest, time to first subsequent arrest, and severity of arrest charges in the 12 months following release were examined for: 1) combined IM + PN and IM groups compared to ETAU; and 2) IM + PN compared to IM.
Results: Within 12 months of release from the index incarceration, 50.5% of the sample had been arrested. The majority of arrest charges (71%) were for low-level, nonviolent crimes. On an intention-to-treat basis, there were no significant differences between the combined IM + PN and IM groups *vs.* ETAU or IM + PN *vs.* IM in the likelihood of arrest, time to first subsequent arrest, or severity of arrest charges.
Conclusion: Initiating IM with or without PN during pretrial detention did not have a significant effect on subsequent arrest during a 12-month post-release follow-up compared to not starting methadone maintenance during detention, despite the high rate of methadone treatment entry in the community following release. This finding may be attributable to the considerable attrition from treatment in the community or other systematic factors. Additional interventions may be needed to reduce the likelihood of subsequent arrest.

1. Introduction

From June 30, 2016, to June 30, 2017, there were 10.6 million admissions to U.S. jails, which held an estimated 745,200 individuals at the end of June 2017 (Zeng, 2019). Opioid use disorder (OUD) is prevalent among jail detainees (Boutwell, Nijhawan, Zaller, & Rich, 2007; Bronson, Stroop, Zimmer, & Berzofsky, 2017), yet most U.S. jails do not offer treatment with medications (Lee & Rich, 2012; Nunn et al., 2009). Following release, detainees with OUD are at high risk of relapse and overdose (Binswanger, Blatchford, Mueller, & Stern, 2013; Bird & Hutchinson, 2003; Bukten et al., 2017; Degenhardt et al., 2016; Farrell & Marsden, 2008). Additionally, this population has a heightened risk of subsequent arrest and incarceration for new crimes as well as for technical violations resulting from failure to comply with the terms of community supervision (Campbell, 2014; Phelps, 2013).

In community-based programs, methadone maintenance treatment (MMT) is associated with decreased illicit opioid use, drug injection, and overdose death (MacArthur et al., 2012; Mattick, Breen, Kimber, & Davoli, 2009). However, there have been mixed findings from evaluations of community-based MMT's impact on self-reported criminal behavior (Ball & Ross, 1991; Hubbard et al., 1989; Hubbard, Craddock, Flynn, Anderson, & Etheridge, 1997; Mattick et al., 2009) and on arrests as obtained from official records (Bowden, Maddux, & Esquivel, 1978; Newman, Baskow, & Cates, 1973; Rothbard et al., 1999; Schwartz et al., 2017). These mixed findings could be attributed to limitations in observational studies, such as lack of control groups, selection bias, regression to the mean, and loss to follow-up (for those studies involving participant interviews). In one of the few studies with a no-treatment control group, Schwartz et al. (2009) found that newly admitted MMT patients in community treatment programs assigned to begin methadone without counseling (termed interim methadone [IM]) had significantly fewer arrests than participants in a control group assigned to a waiting list at 6 months (but not at 12 months) following study enrollment (Schwartz et al., 2009).

Abbreviations: IM + PN, Interim Methadone + Patient Navigation; IM, Interim Methadone; ETAU, enhanced treatment-as-usual; OUD, opioid use disorder; MMT, methadone maintenance treatment; ICC, intraclass correlation coefficient

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Findings from community-based MMT studies should not be generalized to studies of initiating MMT in jail because individuals in jail may have different characteristics, including more severe criminal justice and drug use profiles, than individuals entering treatment in the community (Schwartz, Kelly, Mitchell, Gryczynski, et al., 2019). There have been relatively few U.S. studies of out-of-treatment individuals initiating MMT in jail. Such studies include three retrospective reports on MMT (Bellin et al., 1999; Magura, Rosenblum, Lewis, & Joseph, 1993; Tomasino, Swanson, Nolan, & Shuman, 2001) and two randomized trials (Dole et al., 1969; Magura et al., 2009), all conducted at Rikers Island Jail in New York City. Using official arrest data from 1996 and 1997, Bellin et al. (1999) found that inmates released on a "high" methadone dose (≥60 mg) had a lower risk of subsequent incarceration than inmates on a "low" dose (≤30 mg) as well as a significantly longer time to first subsequent arrest (253 *vs.* 187 days). Surprisingly, high-dose inmates had a significantly higher risk of incarceration than inmates who received a methadone detoxification or no treatment at all, possibly due to the longer history of criminal behavior and more chronic nature of OUD of the high-dose participants. However, the study did not use random assignment nor make it clear whether patients had continued to receive the methadone dose with which they were treated in community treatment prior to arrest, were out of treatment at the time of arrest, or both. In a pilot study, Dole et al. (1969) found that 3 of the 12 (25%) randomly selected short-term sentenced inmates who started methadone treatment prior to release were subsequently arrested during the 7-to-10-month follow-up period in contrast to 15 of the 16 (94%) inmates who were not able to start methadone treatment in jail. Magura et al. (2009) found no difference in the mean number of self-reported arrests in their 3-month post-release follow-up between groups randomly assigned to start either buprenorphine or methadone maintenance prior to release (mean [standard deviation] arrests = 0.69 [0.95] for buprenorphine and 0.71 [0.77] for methadone). This latter study did not have a no medication control group.

Our research team previously reported on a randomized clinical trial in the Baltimore City Detention Center among newly arrested detainees with OUD who were being treated for opioid withdrawal and were not enrolled in opioid agonist treatment at the time of the index arrest (Schwartz et al., 2016; Schwartz, Kelly, Mitchell, O'Grady, et al., 2019). The study examined the effectiveness of initiating interim methadone maintenance treatment with PN during pretrial detention compared to interim methadone maintenance treatment without PN, compared to a brief, medically supervised withdrawal with a referral to community treatment (see Schwartz, Kelly, Mitchell, O'Grady, et al., 2019). We previously reported on the primary outcomes (entry into treatment 1 month after release and opioid-positive drug tests) and select secondary outcomes (*e.g.*, treatment retention, self-reported opioid and cocaine use). In the current analysis, we examine the impact of the study interventions on likelihood of subsequent arrest, time to first arrest, and the severity of arrest charges during the 12 months after release.

2. Materials and methods

2.1. Parent study

This was a three-group randomized clinical trial in which newly arrested male and female patients (N = 225) receiving medically managed opioid withdrawal treatment in the Baltimore City Detention Center enrolled in the study from December 2014 through October 2017. For the duration of the study, methadone treatment was available to some individuals who were not study participants to medically manage withdrawal or to continue methadone maintenance treatment if they had been enrolled in treatment at the time of arrest. Only pregnant women were routinely initiated on methadone maintenance treatment in the detention center. Buprenorphine and naltrexone treatment were not available to detainees.

Study inclusion criteria were: 1) meeting DSM-5 OUD criteria; 2) being detained for at least 48 h for a charge likely to result in a sentence of less than one year (excluding violent crimes such as armed robbery, arson, rape, and murder); 3) being treated for opioid withdrawal; 4) being able and willing to provide informed consent in English; and 5) planning to live in the Baltimore area upon release. Individuals were excluded if they were: 1) receiving methadone or buprenorphine treatment in the community; 2) medically or psychiatrically unstable; 3) pregnant; or 4) receiving treatment for moderate or severe alcohol and/or sedative hypnotic withdrawal. All participants provided informed consent. The study was approved by the Friends Research Institute Institutional Review Board (IRB) and the Western IRB.

The detention center's medical staff referred willing patients who were receiving medically supervised opioid withdrawal treatment to the research assistants (RAs) for study eligibility screening and to provide informed consent. RAs administered the baseline study assessments and obtained the patient's medical record for review by the principal investigator (PI). The RAs then assigned eligible participants to group, within gender, in blocks of 3, 6, or 9 by opening the next opaque, sealed, and numbered randomization envelope that was created by the study's statistician using a random permutation procedure.

Participants were randomly assigned to: 1) interim methadone with patient navigation (IM + PN; n = 75); 2) IM without PN (IM; n = 74); or 3) enhanced treatment-as-usual (ETAU; n = 76). IM refers to methadone treatment without routine counseling when such counseling is not available. Community studies have shown IM to increase treatment entry and to reduce illicit opioid use (Schwartz et al., 2006; Yancovitz et al., 1991). IM consisted of MMT without counseling during incarceration and continuation of methadone maintenance with counseling upon release at one of four participating community programs. IM + PN participants received IM and continuation of methadone maintenance with counseling upon release at one of the four programs plus three months of patient navigation services (Metsch et al., 2016; Sorensen et al., 2005). A dedicated staff person provided PN to facilitate a smooth transition to community MMT entry after release. The patient navigator had previously worked in that capacity on a NIDA-funded study (Metsch et al., 2016). She was trained for the current study using a strengths-based case management manual based on prior work (Metsch et al., 2016; Sorensen et al., 2005). Participants assigned to the IM + PN group met with the navigator once prior to release to develop a post-release plan for entering and remaining in treatment. She was available to work with participants over a 3-month period following release to address barriers to entering and receiving OUD treatment and to help meet basic needs such as obtaining ID cards and bus passes using a small fund that was available through the study. ETAU consisted of a brief medically supervised withdrawal with methadone and the provision of drug education/overdose prevention information and the community treatment access hotline number.

Follow-up interviews were conducted at 1, 3, 6, and 12 months following release from incarceration. Interviews included, among other measures, the Addiction Severity Index (McLellan et al., 1992), questions assessing treatment enrollment status, and a urine drug test. Participants were paid $30 for each follow-up (but not for the baseline interview). More information on study methods may be found at Schwartz et al. (2016) and Schwartz, Kelly, Mitchell, O'Grady, et al. (2019).

2.2. Current study procedures

Of the 225 participants in the parent study, 212 were released in time to obtain the full 12 months of post-release arrest data. Research staff searched the Maryland Judiciary Case Search (MJCS) website (http://casesearch.courts.state.md.us/casesearch/inquirySearchParam.jis) and obtained dates of arrest and arrest charges for any arrests that occurred during the 12-month period following initial release

(including new crimes and violations of probation). MJCS is a public online database that provides access to detailed case information for Maryland circuit and district court cases. Data on arrests for violations of parole were not available on the case search website and, hence, could not be included for the 11 participants who reported being on parole at baseline. Arrest data were gathered both for participants released directly to the community from detention and for participants who were sentenced, tapered off of methadone (as methadone is not provided to inmates in prison facilities in Maryland), and transferred to prison to serve their sentence prior to their release to the community.

The study's PI (RPS), co-investigators (JHJ and SMK), and RA (TND) independently rated arrest charges from 1 (least severe) to 7 (most severe) using a scale that Nurco and colleagues (Nurco, Hanlon, Balter, Kinlock, & Slaght, 1991) developed and that Schwartz et al. (2009, 2017) adapted. Crimes involving violence or the infliction of physical harm were rated the most severely (*e.g.*, murder, rape, assault), followed by crimes involving loss or destruction of property (*e.g.*, theft). Crimes in which there was no direct victim (*e.g.*, simple drug possession, prostitution) were given the lowest severity ratings. Interrater reliability (intraclass correlation coefficient [*ICC*](2,*k*); Shrout & Fleiss, 1979) was calculated. The results of the ICC indicate that there was high agreement among the raters [$ICC(2,4) = 0.93$] and that the mean of the ratings could be used as the measure of severity.

2.2.1. Outcome variable

The outcome measure was arrest (yes/no) during the 12-month period following release.

2.2.2. Explanatory variables

Explanatory variables used in the current analysis are the same as those used in the prior report (Schwartz, Kelly, Mitchell, O'Grady, et al., 2019). These included several baseline variables obtained from the study's demographic measure and the Addiction Severity Index (McLellan et al., 1992), including gender, age, self-reported number of days of cocaine use in the past 30 days, and previous methadone maintenance treatment enrollment (yes/no). Treatment group was included as an explanatory variable, with the focus of the analysis on specific subeffects within the treatment group effect—two orthogonal, single-degree-of-freedom planned contrasts: 1) combined IM + PN and IM groups compared to ETAU; and 2) IM + PN compared to IM. We also included a main effect for gender and a treatment group x gender interaction, because we believed there was sufficient prior research to indicate that gender could serve as a moderator of treatment success (Hser, Huang, Teruya, & Anglin, 2004; Magura et al., 1993)

2.2.3. Statistical analysis

As in the prior report (Schwartz, Kelly, Mitchell, O'Grady, et al., 2019), analyses were conducted on an intention-to-treat (ITT) basis as well as a modified intention-to-treat (MITT) basis. The ITT analysis ($N = 212$) included the 197 participants who were released to the community directly from the index detention as well as the 15 participants who had been sentenced during the index detention, had their methadone discontinued (for participants in IM + PN or IM groups only), served time in prison, and were released in time to complete a 12-month follow-up period. The modified ITT analysis was conducted including only the 197 participants who were released directly to the community from the detention center (excluding the 15 participants who had been sentenced and served time in prison prior to release). All analyses were conducted in SPSS version 25. Logistic regression was utilized to examine the factors that predicted the likelihood of arrest in the 12-month follow-up period under the assumption that this outcome variable followed a binomial distribution.

2.2.4. Supplemental analyses

Several supplemental analyses of interest were conducted with both the ITT and the MITT samples. First, Cox proportional hazards regression was utilized to predict time to first subsequent arrest (censored at 365 days) in the 12-month follow-up period. Second, severity of the most severe arrest charge was examined using a generalized linear model approach coding severity as a continuous measure ranging from 0 (no arrest during follow-up) to 7 (most severe charge), assuming a negative binomial distribution for severity rating. All analyses included the same explanatory variables and planned contrasts listed above (see Section 2.2.2.) as fixed effects in the model. Finally, because many individuals are arrested solely for probation violations resulting from technical offenses (*e.g.*, missing appointments with a probation agent or testing positive for drugs or alcohol; Phelps, 2013), we conducted the same analyses assessing arrest, severity, and time to first arrest with the ITT sample and the MITT sample but did not count arrests for violations of probation.

3. Results

3.1. Participants

The sample of 212 participants in the current analysis primarily consisted of men ($n = 168$; 79.2%) and the majority were African American ($n = 132$; 62.3%); mean (*SD*) age was 38.0 (10.4) years. Participants reported a mean (*SD*) of 66.5 (71.9) lifetime months of incarceration at enrollment, and nearly half (46.7%) were on parole (5.2%) or probation (41.5%). Participants reported using heroin for a mean (*SD*) of 28.1 (5.7) of the past 30 days at baseline, and reported using cocaine for 15.2 (13.6) days. Just over a third (36.3%) reported a previous enrollment in methadone maintenance treatment. There were no significant differences in baseline demographic, substance use, criminal justice, or treatment variables among study groups ($ps > .05$).

3.2. Likelihood of arrest

3.2.1. ITT analysis

Of the 212 participants, 107 (50.5%) were arrested during the 12 months after release (32 [45.1%] IM + PN participants, 39 [56.5%] IM participants, and 36 [50.0%] ETAU participants; see Table 1). There were 72 (34.0%) participants with 1 arrest, 20 (9.4%) with 2 arrests, 7 (3.3%) with 3 arrests, 6 (2.8%) with 4 arrests, and 2 (0.9%) with 5 arrests.

Table 2 shows the statistical tests for the two planned contrasts and main effects for the likelihood of arrest during follow-up. Neither of the planned contrasts (combined IM + PN and IM group *vs.* ETAU and IM + PN *vs.* IM) was significant ($ps > .05$). Age was the only significant explanatory variable, indicating that as age increased, the likelihood of arrest in the 12-month follow-up period decreased ($p = .006$).

3.2.2. MITT analysis

In the MITT subsample of 197 participants who were released directly to the community from detention (*i.e.*, not sentenced, transferred to prison, and subsequently released from prison), 100 participants (50.8%) were arrested during follow-up. As with the ITT analysis, there were no significant differences in the planned contrasts (combined IM + PN and IM groups *vs.* ETAU and IM + PN *vs.* IM) in the MITT analysis (all $ps > .05$). Also similar to the ITT analysis, age was the only explanatory variable significantly related to the likelihood of arrest ($p = .007$).

3.3. Supplemental analyses

3.3.1. Time to first subsequent arrest

3.3.1.1. ITT analysis. Cox regression also revealed that neither of the two planned contrasts was significant for time to first arrest in the 12-month follow-up period ($ps > .05$; see Table 3). As with arrest status during the 12-month follow-up period, age was a significant

Table 1
Number (%) of arrests in the 12-month post-release follow-up period (N = 212).

Number of arrests	Total sample (N = 212)	IM + PN group (n = 71)	IM group (n = 69)	ETAU group (n = 72)
0	105 (49.5%)	39 (54.9%)	30 (43.5%)	36 (50.0%)
1	72 (34.0%)	24 (33.8%)	26 (37.7%)	22 (30.6%)
2	20 (9.4%)	4 (5.6%)	8 (11.6%)	8 (11.1%)
3	7 (3.3%)	3 (4.2%)	3 (4.3%)	1 (1.4%)
4	6 (2.8%)	1 (1.4%)	2 (2.9%)	3 (4.2%)
5	2 (0.9%)	0 (0.0%)	0 (0.0%)	2 (2.8%)

Note: IM + PN = Interim Methadone + Patient Navigation; IM = Interim Methadone; ETAU = Enhanced Treatment as Usual. Percentages are relative to the sample size in each group. Column percentages may not total to 100% due to rounding.

Table 2
Test of effects for likelihood of arrest in the 12-month post-release follow-up period (N = 212).

	Test statistic (*df*)	*p*
Group	$\chi^2(2) = 0.67$	.72
Planned contrasts:		
Combined IM + PN and IM *vs.* ETAU	$\chi^2(1) = 0.36$	.55
IM + PN *vs.* IM	$\chi^2(1) = 0.32$	.57
Gender (ref. = male)	$\chi^2(1) = 1.8$	.18
Age	$\chi^2(1) = 7.7$	*.006*
Prior methadone maintenance treatment (MMT) enrollment (yes/no; ref. = no)	$\chi^2(1) = 1.5$	.22
Baseline number of days used cocaine (past 30 days)	$\chi^2(1) = 0.71$	.40
Group X Gender	$\chi^2(2) = 0.42$	.81

Note: IM + PN = Interim Methadone + Patient Navigation; IM = Interim Methadone; ETAU = Enhanced Treatment as Usual.

explanatory variable indicating a 3.5% decrease in the expected arrest hazard relative to a one-year increase in age ($p < .001$).

3.3.1.2. MITT analysis. Findings were similar to the ITT analysis with age as the only significant explanatory variable ($p < .001$).

3.3.2. Severity of most severe charge

3.3.2.1. ITT analysis. For the severity analyses, one participant with a charge listed as "other" was excluded resulting in a total sample of 211. A total of 31 (14.7%) participants were arrested for more severe charges rated ≥5 (*i.e.*, crimes potentially involving violence or the infliction of physical harm), such as burglary, assault, robbery, or armed robbery (see Table 4). There were no differences in either the combined IM + PN and IM *vs.* ETAU contrast or the IM + PN *vs.* IM contrast for severity of most severe charge. Both gender (p = .018) and age (p = .003) were significant, indicating that female gender and older age were associated with lower charge severity (see Table 5).

3.3.2.2. MITT analysis. The findings were similar to the ITT analysis, in that only gender (p = .022) and age (p = .003) were significantly related to the severity of arrest charge.

3.3.3. Analyses of arrests that were not solely for violations of probation

3.3.3.1. ITT analysis. Of the ITT sample of 211 (1 participant with arrest charge "other" was excluded) participants, 90 (42.7%) participants were arrested in the 12-month post-release period for charges that were not solely for violations of probation. In terms of the likelihood of arrest, there were no significant differences in the planned contrasts (combined IM + PN and IM group *vs.* ETAU and IM + PN *vs.* IM; all *ps* > .05). Older age (p = .007) and female gender were significantly associated with lower likelihood of arrest (estimated marginal mean [*SE*] likelihood of arrest for males and females was 0.48 [0.05] and 0.25 [0.08], respectively; p = .045).

Similarly, neither planned contrast was significant in the Cox regression of time to first arrest (*ps* > .05). Both age (p = .001) and gender (p = .013) were significant predictors, with older age and female gender again indicating lower arrest hazard. Finally, neither planned contrast was significant in the analysis of the most severe charge (*ps* > .05), with female gender (p = .002), older age (p = .002), and fewer days of cocaine use (p = .041) significantly associated with lower severity.

3.3.3.2. MITT analysis. Eighty-four (42.9%) of the 196 participants in the MITT sample were arrested when violations of probation were not counted as arrests. As with the ITT analyses, neither planned contrast was significant for likelihood of arrest, time to first arrest, or severity of arrest charge (*ps* > .05). For likelihood of arrest, age was again significant (p = .012). For time to first arrest and charge severity, age (*ps* = .001 and .004 for time to first arrest and severity, respectively) and gender (*ps* = .015 and .003 for time to first arrest and severity, respectively) were again significant predictors.

4. Discussion

In this randomized clinical trial, we examined the likelihood of arrest, time to first subsequent arrest, and severity of charges in the 12 months following release from index arrest as obtained from official arrest records for 212 adults enrolled in a study of initiating methadone treatment in jail either with or without patient navigation *vs.* enhanced treatment-as-usual in Baltimore. This is the first randomized clinical trial of initiating methadone in jail with pretrial detainees that includes a contemporaneous control group.

Approximately half of the study sample was arrested during the 12-month follow-up period after release. There were no significant differences found in likelihood of arrest, time to first subsequent arrest, or

Table 3
Hazard ratios, 95% confidence intervals (*CI*), and *p* values for time to first arrest in the 12-month post-release follow-up period (N = 212).

	Hazard ratio	95% *CI* (lower, upper)	*p*
Group			.52
Planned contrasts:			
Combined IM + PN and IM *vs.* ETAU	1.0	0.65, 1.60	.94
IM + PN *vs.* IM	1.4	0.80, 2.31	.25
Gender (ref. = male)	0.61	0.36, 1.05	.07
Age	0.97	0.95, 0.98	*< .001*
Prior methadone maintenance treatment (MMT) enrollment (yes/no; ref. = no)	1.4	0.94, 2.21	.09
Baseline number of days used cocaine (past 30 days)	1.0	0.99, 1.02	.37
Group X Gender			.65
IM + PN and IM *vs.* ETAU X Gender	0.64	0.23, 1.84	.41
IM + PN *vs.* IM X Gender	0.77	0.24, 2.41	.65

Note: IM + PN = Interim Methadone + Patient Navigation; IM = Interim Methadone; ETAU = Enhanced Treatment as Usual. A hazard ratio for the Group effect is not presented because there are three treatment groups and so there is more than one hazard ratio.

Table 4
Severity rating of arrest charges and number (%) of participants in each severity category for most severe charge in the 12-month post-release follow-up period (N = 211[a]).

Severity rating	Examples of criminal charges	*N* (%)
0	No arrest	105 (49.8%)
1	Prostitution; Controlled dangerous substance (CDS) possession; Consuming alcohol in public; Driving vehicle in excess of reasonable speed	12 (5.7%)
2	Trespassing; Failure to obey traffic instructions; Disorderly conduct; Malicious destruction of property-less than $500	21 (10.0%)
3	Theft: less than $500; Malicious destruction of property-more than $500; Forgery-prescription; Credit card theft	9 (4.3%)
4	Theft: $1000–$10,000; Handgun on person; Controlled dangerous substance (CDS) distribution of narcotics	33 (15.6%)
5	Burglary; Assault-2nd degree; Deadly weapon with intent to injure	18 (8.5%)
6	Robbery; Armed robbery; Assault-1st degree; Kidnapping	13 (6.2%)
7	Attempted murder (1st or 2nd degree); Murder (1st or 2nd degree); Rape (1st or 2nd degree)	0 (0.0%)

Note: Percentages may not total to 100% due to rounding.
[a] Severity rating missing for 1 participant for charge listed as "other."

Table 5
Test of effects for severity of most severe charge in the 12-month post-release follow-up period (N = 211).

	Test statistic (*df*)	*p*
Group	$\chi^2(2) = 0.53$	.77
Planned contrasts:		
Combined IM + PN and IM *vs.* ETAU	$\chi^2(1) = 0.17$	.68
IM + PN *vs.* IM	$\chi^2(1) = 0.38$	.54
Gender (ref. = male)	$\chi^2(1) = 5.6$	*.018*
Age	$\chi^2(1) = 8.9$	*.003*
Prior methadone maintenance treatment (MMT) enrollment (yes/no; ref. = no)	$\chi^2(1) = 0.39$	.53
Baseline number of days used cocaine (past 30 days)	$\chi^2(1) = 2.4$	.12
Group X Gender	$\chi^2(2) = 2.3$	.32

Note: IM + PN = Interim Methadone + Patient Navigation; IM = Interim Methadone; ETAU = Enhanced Treatment as Usual. Possible range is 0–7, where 0 = no arrest, 1 = least severe charge, and 7 = most severe charge. Severity rating missing for 1 participant for charge listed as "other."

severity of charges during the 12-month follow-up period when comparing the methadone treatment groups combined *vs.* the enhanced treatment-as-usual group or the methadone group with patient navigation *vs.* the methadone group without patient navigation. Findings remained consistent even when technical violations of probation were not counted as arrests. These results are not unexpected when examined in the context of the previously reported findings from this study, which showed that there were no differences in rates of opioid- and cocaine-positive urine tests and self-reported frequency of heroin use, cocaine use, and criminal activity among treatment groups (Schwartz, Kelly, Mitchell, O'Grady, et al., 2019).

We are not aware of any other randomized clinical trials that compared starting methadone maintenance in jail among newly arrested detainees compared to a group of detainees who did not start methadone maintenance. However, there have been random assignment studies conducted among prisoners with short sentences (Dole et al., 1969) and prisoners with longer sentences who had been withdrawn from opioids during incarceration (Kinlock, Gordon, Schwartz, Fitzgerald, & O'Grady, 2009; McKenzie et al., 2012) and who were inducted slowly on methadone, beginning with a low dose prior to release. In the Dole and colleagues pilot study, inmates started on methadone maintenance (n = 12) prior to release were less likely to be arrested following release during a 7-to-10-month follow-up compared to a control group (n = 16) that did not start methadone prior to release. However, this study was limited by a small sample size and variable follow-up period. McKenzie et al. (2012) found no difference in post-release arrest over a 6-month follow-up period between participants randomly assigned to initiate methadone prior to release (n = 29) *vs.* after release (n = 29). In a three-group randomized clinical trial (Kinlock et al., 2009), there were no significant differences in subsequent arrest over a 12-month follow-up period among groups assigned to begin methadone treatment prior to release (n = 71), begin methadone treatment following release (n = 70), or begin counseling in prison without methadone (n = 70).

Older age was significantly associated with lower arrest rates in this study, as well as longer time to first subsequent arrest and less severe arrest charges during the 12-month follow-up period. These findings were consistent with primary study results that showed a significant inverse association between age and self-reported criminal activity (Schwartz, Kelly, Mitchell, O'Grady, et al., 2019). Two prior longitudinal studies of recidivism among individuals involved in the criminal justice system found that older age was significantly associated with lower rates of recidivism (Olver & Wong, 2015; Sampson & Laub, 2003). Kinlock et al. (2009) found that older age was significantly associated with a lower likelihood of arrest during the 12-month post-release follow-up. Other studies exploring patterns of criminal activity among individuals with substance use disorders have found lower rates of acquisitive crimes (Gossop, Marsden, Stewart, & Rolfe, 2000; Hayhurst et al., 2013; Horyniak et al., 2016) and arrests (Horyniak et al., 2016; Kopak, Lawson, & Hoffmann, 2018) among older individuals.

Women were significantly less likely than men to be charged with a serious crime in the 12-month follow-up period. Additionally, when probation violations were not counted as arrests, women were significantly less likely than men to be arrested after release, less likely to be arrested for a more severe charge, and had a longer time to first subsequent arrest than men. Although community supervision is intended as an alternative to prison, monitoring supervised individuals leaves them at an increased risk of violating the terms of their supervision and returning to jail (Phelps, 2013; Tonry & Lynch, 1996). Individuals may have difficulty complying with the numerous requirements of supervision, such as attending meetings, paying supervision fees, and abstaining from substance use. This may be especially true for women who may have difficulties securing childcare or lack support from partners (Swavola, Riley, & Subramanian, 2016). Additionally, prior research has indicated that women may be subjected to greater supervision than men (Caputo, 2014; Morash, Kashy, Smith, & Cobbina, 2019), leaving them at higher risk of violating their probation.

One strength of the current study was its randomized design with a control group that did not begin methadone maintenance treatment during incarceration. In addition, we also used official arrest records rather than self-report, which allowed us to obtain data on the entire study sample. In contrast to studies on continuing community-based methadone treatment during incarceration (Rich et al., 2015), methadone treatment was initiated for detainees in the current study who were not in treatment at the time of their index arrest. Further, the study population in this report appears to be different from other populations entering treatment in the community. In a previous report, we noted that participants in the parent study sample—compared to a sample of methadone patients entering treatment from the community—were more likely to have been arrested in the year prior to study enrollment (excluding the index arrest) and to have been on parole or

probation at baseline, were less likely to have prior methadone treatment admissions, had more lifetime months of incarceration, and reported using cocaine and drinking to intoxication on more days (Schwartz, Kelly, Mitchell, Gryczynski, et al., 2019).

Our study sample was recruited from a large detention center in Baltimore City and so findings may not generalize to other cities. The study sample was not representative of the entire arrestee population because individuals released on their own recognizance within 48 h of arrest were excluded, along with individuals whose arrest charges were more serious and likely to lead to a long sentence if they were found to be guilty (*e.g.*, homicide and rape). The median methadone dose at release from the detention center was 40 mg and, therefore, it is possible that higher doses at release would have resulted in higher treatment entry and retention rates and/or lower subsequent arrest rates. Despite a high rate of MMT entry following release, there was considerable dropout from community treatment in both IM groups, which may limit any potential treatment effects on arrest rates. This study included a small number of women ($n = 44$) due to their lower rates of arrest relative to men. In addition, the number of arrests is a significant underestimate of the number of crimes actually committed, with studies finding that only about 1% of crimes result in arrest (Ball, Shaffer, & Nurco, 1983). Finally, while we were not able to obtain data on violations of parole, only 11 participants were on parole at the time of random assignment.

5. Conclusion

We found no significant difference in arrest rates between the two groups that were assigned to start interim methadone maintenance prior to release and the group that was assigned to a medically managed withdrawal during incarceration with a referral to treatment in the community. The majority of participants who started interim methadone maintenance during detention received their first methadone dose in the community following release. However, community treatment had considerable attrition, which may have contributed to no significant effect of treatment on subsequent likelihood of arrest, time to first subsequent arrest, or severity of arrest charge. A different array of services—such as contingency management to address concurrent cocaine use, income support to reduce reliance on illegal income, or the use of extended-release buprenorphine or extended-release naltrexone to reduce the adherence burden associated with daily methadone treatment—may increase treatment retention and reduce the likelihood of subsequent arrest.

CRediT authorship contribution statement

Sharon M. Kelly: Supervision, Project administration, Investigation, Data curation, Validation, Formal analysis, Writing - original draft, Writing - review & editing. **Robert P. Schwartz**: Conceptualization, Funding acquisition, Supervision, Project administration, Writing - original draft, Writing - review & editing. **Kevin E. O'Grady**: Conceptualization, Methodology, Data curation, Validation, Formal analysis. **Shannon G. Mitchell**: Writing - review & editing. **Tiffany Duren**: Investigation, Writing - review & editing. **Anjalee Sharma**: Investigation, Writing - review & editing. **Jerome H. Jaffe**: Conceptualization, Writing - original draft, Writing - review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Dr. Schwartz has consulted for Verily Life Sciences. The other authors report no conflicts.

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